A COMPOSITION OF H-PROJECTIVE MORPHISMS IS H-PROJECTIVE

THE STACKS PROJECT AUTHORS

Editorial note. All delivered mathematical text is the author's own, copied from the pinned Stacks Project chapter file pinned-original-source.tex; no statement or proof step was rewritten, reordered or removed.

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Editorial note. The preamble is the chapter's shared Stacks Project preamble, retained as sources/stacks-morphisms/retained-preamble.tex: a line-for-line excerpt of the shared Stacks preamble staged with this snapshot. No macro was invented or redefined, and this card adds no preamble text of its own.

Editorial note. Statement span 10630-10633 and proof span 10635-10655 come from the halfA pool card operations/pool-cards/stacks-morphisms-halfA.json and were re-read in the pinned chapter file before this card was written.

Editorial note. The proof contains a large xymatrix diagram exhibiting the composite as a closed subscheme; it is copied verbatim and the retained chapter preamble loads Xy-pic.

Editorial note. The author's identification of a projective space over a projective space with a product of projective spaces is used as written.

Editorial note. 1 cross-references occur inside the counted spans; they are listed in external_references as navigation pointers only.

Allowed standard prerequisites (editorial).

- The definition of an H-projective morphism as one admitting a closed immersion into a projective space over the target compatible with the structure morphisms. Used at proof lines 10635-10638. Supplies the two closed immersions that the proof composes.
- The Segre embedding of a product of projective spaces over a base into a projective space over that base, cited from the constructions chapter. Cited at proof lines 10651-10653. Turns the product of the two projective spaces into a closed subscheme of a single projective space over the base.
- Base change of a projective space and of a closed immersion, and the identification of a projective space over a projective space with the product of two projective spaces over the base. Used at proof lines 10639-10650. Builds the displayed commutative diagram exhibiting the composite as a closed subscheme of a projective space over the base.

Source reference. Original lemma 01WD: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 2478–2478. Same-chapter reference used in the retained author text at source line(s) 9508, 9689, 10652. It is supplied only as a

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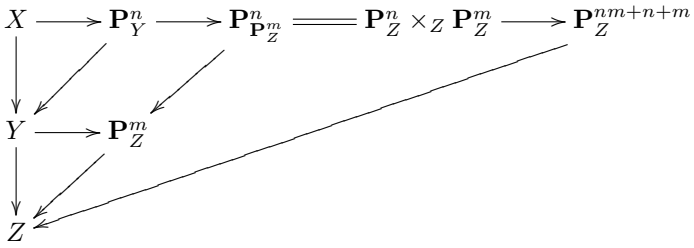
navigation pointer so that the retained text compiles; the counted item delivers the author's own statement and the author's complete proof, and no part of the referenced item is copied in or substituted for the counted argument.

Lemma 0.1. *A composition of H-projective morphisms is H-projective.*

Proof. Suppose $X \to Y$ and $Y \to Z$ are H-projective. Then there exist closed immersions $X \to \mathbf{P}^n_Y$ over Y, and $Y \to \mathbf{P}^m_Z$ over Z. Consider the following diagram



Here the rightmost top horizontal arrow is the Segre embedding, see Constructions, Lemma 01WD. The diagram identifies X as a closed subscheme of $\mathbf{P}^{nm+n+m}_Z$ as desired. □